

RAMCO INSTITUTE OF TECHNOLOGY

(Approved by AICTE, Affiliated by Anna University Chennai)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

REGULATION-2013

Course Outcomes

Semester - I	
Course code and Name	Course Outcomes(CO) After completion of the course, the students will be able to
HS6151 Technical English – I	CO1 Explain clearly, confidently, comprehensibly, and communicate with one or many listeners using appropriate communicative strategies. CO2 Construct cohesively and coherently and flawlessly avoiding grammatical errors, using a wide vocabulary range, organizing their ideas logically on a topic. CO3 Organize different genres of texts adopting various reading strategies. CO4 Distinguish and comprehend different spoken discourses/excerpts in different accents. CO5 Listen to different accents, speeches and presentations.
MA6151 Mathematics	CO1 Infer canonical form from quadratic form using the concepts of Eigen values and Eigen vectors. CO2 Test the convergence of sequences and series. CO3 Apply the techniques of differential calculus to find the evaluate and envelope of curves. CO4 Determine the maxima and minima of functions of two variables using partial derivatives. CO5 Find the area enclosed by plane curves and volume of solids using multiple integrals.
PH6151 Engineering Physics-I	CO1 Describe the crystal structures and various crystal growth techniques. CO2 Analyze the elastic nature of materials and thermal behaviour of materials. CO3 Apply the knowledge of quantum mechanics and classical mechanics in addressing the problems related to science and technology. CO4 Apply the knowledge about designing an auditorium with good acoustical properties and make use of Ultrasonics and its applications in various fields. CO5 Illustrate the advantages of optical communication using LASER.

CY6151 Engineering Chemistry-I	CO1 Describe the methods of polymerization, types, Properties and uses of polymers. CO2 Illustrate the concepts of basic thermodynamics and problem solving skills in various disciplines of Engineering. CO3 Discuss the laws of photochemistry in recognizing the interaction of light with matter and its applications in luminescence and spectroscopy. CO4 Review the use of phase rule in identifying its applications in metallurgy and alloys. CO5 Summarize the basic knowledge in Nanochemistry and distinguish the existing technology with nanotechnology.
GE6151 Computer Programming	CO1 Elaborate the organization of digital computer and design the solution for simple computing problems using algorithm, flowchart and pseudo code. CO2 Apply the different looping structure to solve simple scientific and statistical problems. CO3 Devise the solutions for simple problems using arrays and strings CO4 Demonstrate the usage of dynamic memory allocation and pointer variables. CO5 Illustrate the concepts of structure and union with an example programs.
GE6152 Engineering Graphics	CO1 Sketch the conic sections, special curves, and draw orthographic views from pictorial views and models. CO2 Apply the principles of orthographic projections of points in all quadrants, lines and planes in first quadrant. CO3 Sketch the projections of simple solids like prisms, pyramids, cylinder and cone and obtain the traces of plane figures. CO4 Practice the sectional views of solids like cube, prisms, pyramids, cylinders & cones and extend its lateral surfaces. CO5 Sketch the perspective projection of simple solids, truncated prisms, pyramids, cone and cylinders and sketch the isometric projection of simple machine parts.
GE6161 Computer Practices Laboratory	CO1 Describe the usage of office automation tools CO2 Apply the good programming methods for program development. CO3 Design and implement the C program for simple application CO4 Develop and implement the recursive programs CO5 Implement the C program with the help of Structures and Union

GE6261 Engineering Practices Laboratory	CO1 Construct Electrical and Electronic circuits. CO2 Examine different types of electronic circuits and components. CO3 Recognize electrical safety rules, grounding, general house wiring. CO4 Explore soldering practices.
GE6163 Physics and Chemistry Laboratory - I	CO1 The hands on exercises undergone by the students will help them to apply physics principles of optics and thermal physics to evaluate engineering properties of materials. CO2 Perform the quantitative chemical analysis of chloride and dissolved oxygen. CO3 Determine the amount of acids by using the instruments of conductivity meter and pH meter.
Semester II	
HS6251 Technical English – II	CO1 Speak convincingly, express their opinions clearly, initiate a discussion, negotiate, argue using appropriate communicative strategies.. CO2 Write effectively and persuasively and produce different types of writing such as narration, description, exposition and argument as well as creative, critical, analytical and evaluative writing. CO3 Read different genres of texts, infer implied meanings and critically analyse and evaluate them for ideas as well as for method of presentation. CO4 Listen/view and comprehend different spoken excerpts critically and infer unspoken and implied meanings. CO5 Read and write effectively for a variety of professional and social settings
MA6251 Mathematics – II	CO1 Apply the knowledge of vector calculus in engineering disciplines. CO2 Solve ordinary differential equations that model the engineering problems. CO3 Find the Laplace transform of functions and solve the ordinary differential equations using Laplace transform. CO4 Construct analytic functions and apply the knowledge of conformal mappings in engineering disciplines. CO5 Evaluate contour integration and apply it in engineering problems.

PH6251 Engineering Physics – II	CO1 Describe the conducting materials and their properties. CO2 Analyze the semiconductors and able to differentiate various types of semiconductors. CO3 Apply the knowledge of magnetic and superconducting materials for modern day to day applications. CO4 Explain the properties and applications of dielectrics. CO5 Apply the knowledge about the modern engineering materials for various applications.
CY6251 Engineering Chemistry – II	CO1 Describe water technology in the purification of water in domestic and industrial applications. CO2 Explain the principles of electrochemistry, the factors affecting corrosion and the prevention of corrosion. CO3 Classify the different alternative sources of energy and the generation processes. CO4 Enumerate the different types of engineering materials and their applications. CO5 Discuss the industrial techniques of petroleum processing and the determination of calorific values and combustion parameters.
CS6201 Digital Principles and System Design	CO1 Analyze different methods used for simplification of Boolean expressions CO2 Use Boolean simplification techniques to design a combinational hardware circuit and write HDL code for circuits. CO3 Design and analysis of a given synchronous sequential circuit. CO4 Implement an asynchronous sequential circuit design. CO5 Design digital circuits using PLD
CS6202 Programming and Data Structures I	CO1 Summarize the concept of arrays and pointers in C language. CO2 Illustrate the process of handling files and heterogeneous data types using C programs. CO3 Explain the operations of Abstract Data Type-Linked List with examples. CO4 Apply the operations of Abstract Data Types-Stack and Queue with examples. CO5 Analyze different sorting, searching algorithms and hashing techniques.
GE6262 Physics and Chemistry Laboratory - II	CO1 Ability to test materials by using their knowledge of applied physics principles in optics and properties of matter. CO2 Determine the hardness, alkalinity and metal ion content in the water samples by volumetric titration. CO3 Estimate the water quality parameters by potentiometer, conductometer and flame photometer.

CS6211 Digital Laboratory	CO1 Apply Boolean simplification techniques to design a combinational hardware circuit CO2 Design and Implement combinational and sequential circuits. CO3 Analyze a given digital circuit – combinational and sequential. CO4 Design the different functional units in a digital computer system. CO5 Design and Implement a simple digital system.
CS6212 Programming and Data Structures Laboratory I	CO1 Design and implement C programs for implementing stacks, queues, and linked lists. CO2 Apply good programming design methods for program development. CO3 Apply the different data structures for implementing solutions to practical problems. CO4 Develop searching and sorting programs.
Semester III	
MA6351 Transforms and Partial Differential Equations	CO1 Solve the Partial Differential Equations. CO2 Determine the Fourier series expansion of functions and hence evaluate the value of infinite series. CO3 Apply the method of separation of variables to solve one dimensional wave equation, one dimensional heat equation and two dimensional heat equation. CO4 Find the Fourier transform of functions and also evaluate definite integrals using Fourier transform. CO5 Calculate the Z-transform of discrete time systems and solve the difference equations using Z-transform.
CS6301 Programming and Data Structure II	CO1 Explain the basic concepts of Object Oriented programming. CO2 Apply the concepts of polymorphism, inheritance and virtual functions for problem solutions. CO3 Explore the generic problem solution, standard libraries with required errors by means of exception handling. CO4 Comprehend the usage of different advanced nonlinear data structure - Set, Heaps and Height balanced trees. CO5 Apply the non-linear data structure graph in solving the real world problems.
CS6302 Database Management Systems	CO1 Explore the basic concepts of Database Management system. CO2 Create database using query languages. CO3 Explain the concepts of transaction processing and concurrency control. CO4 Explore and gain the knowledge on internal storage Structure and indexing techniques. CO5 Relate security concepts to databases.

CS6303 Computer Architecture	CO1 Describe the operations and Instructions in “Microprocessor without Interlocked Pipeline Stages” (MIPS) architecture. CO2 Model arithmetic and logic unit including Floating Point Multiplication and Division Algorithms. CO3 Develop MIPS architecture by building pipelined data path and control path. CO4 Analyze pipelined control units with Instruction Level Parallelism. CO5 Classify the performance of different Memory and Input-Output systems.
CS6304 Analog and Digital Communication	CO1 Apply various analog communication techniques in all communication systems. CO2 Apply various digital communication techniques in all communication systems. CO3 Use data and pulse modulation techniques for lighting applications. CO4 Apply Source and Error control coding in both wired and wireless communication systems. CO5 Utilize of multi-user radio communication systems.
GE6351 Environmental Science and Engineering	CO1 Summarize the importance of public awareness on environment and nature of biodiversity. CO2 Describe the various causes, effect and control measures of environmental pollution. CO3 Discuss the human development that leads to environmental disasters, the value of natural resources and their conservation. CO4 Explain the value of public participation in environmental protection, Environmental Management and Legislation Acts and sustainable development. CO5 Review the problems related to population and their remedial measures.
CS6311 Programming and Data Structure Laboratory II	CO1 Develop C++ programs using the concepts of abstraction, encapsulation, constructor, polymorphism, overloading and inheritance for solving problems. CO2 Design and implement C++ programs for manipulating stacks, queues, linked list, trees and graphs. CO3 Apply the different data structures for realizing solutions to practical problems. CO4 Develop recursive programs using trees and graphs.
CS6312 Database Management Systems Laboratory	CO1 Design and implement a database schema for a given problem-domain. CO2 Populate and query a database. CO3 Build and maintain tables using PL/SQL. CO4 Implement the database triggers and functions.

Semester IV	
MA6453 Probability and Queuing Theory	CO1 Apply the knowledge of probability distributions to tackle real life problems, in particular, analyzing the performance of computer systems. CO2 Model and analyze two dimensional random variable problems. CO3 Characterize phenomenon which evolve with respect to time in a probabilistic manner. CO4 Characterize the Markovian queueing system. CO5 Apply the knowledge of Non-Markovian queueing models and queueing networks in solving problems in computer science engineering.
CS6551 Computer Networks	CO1 Describe the Network fundamentals and terminology CO2 Recognize the different internetworking devices and their functions CO3 Explore the network with routing and multicasting CO4 Explain the detailed inner workings of TCP/IP protocol suit. CO5 Analyze the features and operations of various application layer protocols such as HTTP, DNS, and SMTP.
CS6401 Operating Systems	CO1 Summarize the basic concepts, System call, structure and functions of Operating Systems. CO2 Design the various Scheduling algorithms, Deadlock prevention, Deadlock avoidance algorithms and apply the principles of concurrency. CO3 Demonstrate the usage of various memory management schemes. CO4 Encapsulate the concepts of Mass Storage Structure, File System Structure and I/O Systems. CO5 Implement administrative tasks on Linux servers.
CS6402 Design and Analysis of Algorithms	CO1 Describe the fundamentals of algorithmic problem solving and able to analyse recursive and non-recursive algorithms. CO2 Design algorithms for various computing problems using brute force and divide-and conquer technique. CO3 Analyze the time and space complexity of various algorithms using dynamic programming and greedy technique. CO4 Analyze the different algorithm design techniques for a given problem using iterative improvement. CO5 Modify existing algorithms to improve efficiency.
EC6504 Microprocessor and Microcontroller	CO1 Design and implement programs on 8086 microprocessor. CO2 Implement the system bus structure of 8086 and coprocessor. CO3 Describe the I/O devices, peripherals and bus interfacing. CO4 Elaborate the operation of 8051 microcontroller architecture and implement ALP using 8051 instructions. CO5 Design and implement 8051 microcontroller based systems.

CS6403 Software Engineering	CO1 Describe the purpose and facts of different software development process models with an insight into generic process framework. CO2 Identify the functional and non-functional requirements for software development by preparing IEEE Software Requirements Document. CO3 Express software design activities using data flow diagrams and architectural diagrams. CO4 Develop a testing framework by understanding the purposes and stages of software testing and test-driven development. CO5 Explain the project management activities involved in developing a framework including planning, scheduling, risk assessment/management.
CS6411 Networks Laboratory	CO1 Describe the usage of socket programming and client server model. CO2 Implement the different protocols and network commands. CO3 Design and implement the application using TCP concepts. CO4 Implement the algorithms with the help of Network Simulator.
CS6412 Microprocessor and Microcontroller Laboratory	CO1 Write ALP Programs for fixed and Floating Point and Arithmetic. CO2 Interface different I/O with 8086 processor. CO3 Generate waveforms using 8086 processors. CO4 Write and Execute ALP Programs in 8051. CO5 Explain the difference between Simulator and Emulator
CS6413 Operating Systems Laboratory	CO1 Use the basics of shell programming. CO2 Use the System calls and implement in C programming. CO3 Apply the file system related system calls. CO4 Create processes and implement IPC. CO5 Compare the performance of various CPU Scheduling Algorithm, Implement deadlock avoidance, and Detection Algorithms.
Semester V	
MA6566 Discrete Mathematics	CO1 Describe the concepts needed to test the logic of a program. CO2 Identify the structures on many levels and be aware of the counting principles. CO3 Explain graph terminology and special types of graphs. CO4 Illustrate the concepts and properties of algebraic structures such as groups, rings and fields CO5 Explain the concepts of Lattices and Boolean algebra.

CS6501 Internet Programming	CO1 Implement Java programs. CO2 Create a basic website using HTML and Cascading Style Sheets. CO3 Design and implement client side programs using JavaScript and server side programs using Servlets and JSP. CO4 Design and implement simple web page in PHP, and to present data in XML format. CO5 Design rich client presentation using AJAX and implement web services.
CS6502 Object Oriented Analysis and Design	CO1 Use the UML analysis and design diagrams. CO2 Interpret and use the GRASP design patterns and GoF Design patterns. CO3 Analyze and design use case modeling and domain modeling. CO4 Apply appropriate design patterns. CO5 Design and implement projects using Object Oriented concepts and compare various testing techniques.
CS6503 Theory of Computation	CO1 Construct a minimized finite automaton to recognize a given regular language. CO2 Describe formal relationships among machines, languages and grammars. CO3 Construct the pushdown automata for all the context free language. CO4 Discuss the basic properties of Turing Machines and Techniques for turing machine construction. CO5 Explain the decidability or Un-decidability of various problems.
CS6504 Computer Graphics	CO1 Describe the graphics hardware devices, software used and different drawing algorithms. CO2 Apply two dimensional transformations and clipping techniques to graphical objects. CO3 Design three-dimensional graphical objects and apply three-dimensional transformations into graphical objects. CO4 Explain the illumination and color models. CO5 Design an animation sequences.
CS6511 Case Tools Laboratory	CO1 Design and implement projects using Object Oriented concepts. CO2 Use the UML analysis and design diagrams. CO3 Apply appropriate design patterns CO4 Create code from design CO5 Compare and contrast various testing techniques.
CS6512 Internet Programming Laboratory	CO1 Design user interfaces using Java frames and applets. CO2 Design Web pages using HTML/XML and style sheets. CO3 Develop dynamic web pages using server side scripting and write client server applications. CO4 Use the frameworks like JSP Strut, Hibernate, Spring. CO5 Create applications with AJAX and web services.

CS6513 Computer Graphics Laboratory	CO1 Explain the basics of graphics programming. CO2 Create 2D animations. CO3 Implement image manipulation and enhancement. CO4 Create 3D graphical scenes using open graphics library suits.
Semester VI	
CS6601 Distributed Systems	CO1 Describe the trends and challenges in distributed system CO2 Apply network virtualization, remote method invocation and objects. CO3 Demonstrate peer-to-peer services and distributed file system. CO4 Analyze the issues related to scalability, synchronization, transaction processing, concurrency and reliability in distributed system. CO5 Design process and resource management systems.
IT6601 Mobile Computing	CO1 Describe the basic concepts of mobile computing and MAC protocol. CO2 Choose the required functionality at each layer for given application. CO3 Explain the basics of mobile telecommunication systems. CO4 Design Ad hoc networks. CO5 Develop a mobile application.
CS6660 Compiler Design	CO1 Summarize the basic concepts of compiler and its phases. CO2 Implement the functionalities of lexical analysis phase like conversion of regular expression to DFA and minimization of DFA. CO3 Design the parsing table using different parsing techniques and different compiler construction tools. CO4 Explain the translation process and run time environment issues. CO5 Apply the various optimization techniques for effectively generating machine code.
IT6502 Digital Signal Processing	CO1 Analyse the properties of discrete time signal and properties of systems using Z-transform. CO2 Apply the concepts of frequency transformations like DFT, FFT and DCT in analysis of various signals and systems CO3 Design Infinite Impulse response (IIR) digital filters. CO4 Design Finite Impulse response (FIR) digital filters. CO5 Analyse the finite Word length effects in digital filters.
CS6659 Artificial Intelligence	CO1 Identify appropriate AI methods to solve a problem using search technique. CO2 Demonstrate the knowledge in predicate and propositional logic and their roles in logic programming. CO3 Formalise a given problem in the language / framework of different AI methods. CO4 Apply the machine learning techniques in solving the real-world problems. CO5 Elucidate the idea of Knowledge Acquisition and Expert Systems

CS6001 C# and .Net programming	CO1 Describe the major elements of .NET Framework, C# language, and develop programs in using C# on .NET. CO2 Comprehend the usage of Inheritance, Interfaces, Operator overloading, abstract class and exception for problem Solving. CO3 Design, Debug, compile and run an application with database connectivity using ADO .NET. CO4 Design and develop Web based applications on NET. CO5 Explain the concepts of CLR and .NET framework
IT6702 Data Warehousing and Data Mining	CO1 Summarize the steps in building a Data Warehouse Architecture and schemas for Decision support. CO2 Explain the usage of OLAP tools used for Querying and Reporting in Business Analysis. CO3 Explain the functionalities of a Data Mining System and its classification. CO4 Apply the Association Rule mining, Classification and Prediction Techniques on large datasets. CO5 Summarize the major clustering methods and applications of data mining techniques in Knowledge discovery.
CS6611 Mobile Application Development Laboratory	CO1 Explain the architecture of mobile application development frameworks. CO2 Choose the required architecture based upon the mobile application to be developed. CO3 Design mobile applications using various layout and widgets. CO4 Implement various mobile applications using emulators. CO5 Deploy applications to hand-held devices.
CS6612 Compiler Laboratory	CO1 Explain and Use the compiler writing tools. CO2 Implement the different Phases of compiler using tools. CO3 Analyze the control flow and data flow of a typical program. CO4 Optimize a given program. CO5 Generate an assembly language program equivalent to a source language program.
GE6674 Communication and Soft Skills - Laboratory Based	CO1 Identify and interpret visuals, communicate in formal and informal conversations, give presentations, and participate in GD. CO2 Explain reading comprehension passages of higher levels, draft Resume, cover letter, reports, emails, and write blogs. CO3 Differentiate between IELTS & TOEFL and take placement oriented verbal ability tests. CO4 Demonstrate appropriate verbal, non-verbal and paralinguistic
Semester VII	
CS6701 Cryptography and Network Security	CO1 Interpret the basic concepts, OSI security architecture, finite fields and number theory. CO2 Compare the various Cryptographic techniques. CO3 Determine the usage of hash functions and digital signature. CO4 Design the various secure applications. CO5 Inject secure coding in the developed applications.

CS6702 Graph Theory and Applications	CO1 Write precise and accurate mathematical definitions of objects in graph theory. CO2 Use mathematical definitions to identify and construct examples of spanning trees and planar graphs. CO3 Validate and critically assess a mathematical proof in graphs and digraphs. CO4 Apply the techniques of permutations and combinations and Binomial theorem for solving problems in Engineering CO5 Construct and solve generating functions, homogeneous and non-homogeneous recurrence relations.
CS6703 Grid and Cloud Computing	CO1 Describe the architecture of Grid and Cloud Computing. CO2 Apply the knowledge to solve the large-scale problem in grid computing. CO3 Explore the concepts of Virtualization. CO4 Develop the web services using the grid and cloud technologies. CO5 Apply security mechanism in grid and cloud computing.
CS6704 Resource Management Techniques	CO1 Apply the knowledge of linear programming problems in engineering disciplines CO2 Solve LPP using dual simplex method, transportation and assignment problems CO3 Apply integer programming to solve real life problems CO4 Solve problems in classical optimization theory CO5 Use PERT and CPM for problems in project management.
CS6005 Advanced Database Systems (E-II)	CO1 Explain the basic foundations of parallel and distributed databases. CO2 Summarize the design principles of active database. CO3 Use query languages and design the different types of database. CO4 Explain the logic and rules of complex queries. CO5 Explore and gain the knowledge on indexing techniques and importance of text and multimedia databases.
IT6801 Service Oriented Architecture (E-II)	CO1 Develop a simple XML document coding and XML Schema. CO2 Create an application based on XML and database. CO3 Compare the characteristics and principles of Service oriented architecture with client server and distributed architecture. CO4 Describe the web services using WSDL, SOAP and UDDI. CO5 Build a Service oriented architecture based applications for intra-enterprise and inter-enterprise applications using J2EE.

IT6006 Data Analytics (E-III)	CO1 Apply the statistical analysis methods. CO2 Compare and contrast various soft computing frameworks and learn efficient algorithms for mining the data from large volumes. CO3 Apply Stream data model. CO4 Perform various association mining and clustering techniques for extracting information from data. CO5 Design distributed file systems and perform visualization.
CS6711 Security Laboratory	CO1 Implement the cipher techniques CO2 Develop the various security algorithms CO3 Use different open source tools for network security and analysis.
CS6712 Grid and Cloud Computing Laboratory	CO1 Develop application using Globus toolkit. CO2 Develop web services/application using Grid framework CO3 Run Virtual machine and install software on it. CO4 Setup a private cloud using OpenStack / Open Nebula / Ecualyptus CO5 Implement applications using MapReduce approach in Hadoop environment.
Semester VIII	
CS6801 Multi –Core Architectures and Programming	CO1 Describe the parallel architecture and parallel programming model CO2 Analyze the issues related to various challenges in parallel programming CO3 Develop parallel programming applications using openMP CO4 Design and develop distributed programming application using openMPI. CO5 Compare and analyze the programming model for serial processor and parallel processor implementation.
CS6008 Human Computer Interaction (E-IV)	CO1 Explain the basic foundations of Human Computer Interaction. CO2 Design effective HCI for individuals and persons with disabilities. CO3 Simplify the issues in the HCI Models and assess the importance of user feedback CO4 State the Mobile HCI implications for designing multimedia/ ecommerce/ e-learning Web sites. CO5 Develop the meaningful user interface.
GE6075 Professional Ethics in Engineering	CO1 Apply ethics, morals and human values in society. CO2 Explain about engineering ethics. CO3 Describe the responsibilities of engineers as experimenters. CO4 Recognize the safety, risks, risk benefit analysis and rights of an engineer. CO5 Discuss the importance of the global issues, moral leadership and code of conduct.

CS6811 PROJECT WORK	CO1 Apply the fundamental knowledge and skills, Which are acquired within the technical area, to a given problem CO2 Identify and summarize an appropriate list of literature review, analyze previous researchers' Work and relate them to the project. Within given constraints, even with limited information, the students will be able to independently analyze and discuss complex inquiries/problems and handle larger problems on the advanced level within the technical area. CO3 Design engineering solutions to complex problems in a systematic approach. Identify and apply appropriate parameters, assumptions and design criteria in consideration of health and safety (example: the use of codes of practice), ethics, economics, environment, sustainability. CO4 Apply research and conduct experiments, as well as to analyze and interpret data that yield the results and answer important applicable research questions. CO5 Utilize technology tools for communication, collaboration, information management, and decision support. CO6 Demonstrate the knowledge, skills and attitudes of a professional engineer. CO7 Interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment. CO 8 Demonstrate a strong working knowledge of ethics and professional responsibility. CO 9 Document and present one's own work, for a given target group, with strict requirements on structure, format, and language usage. CO 10 Present the project outlining the approach and expected results using good oral and written presentation skills. CO 11 Demonstrate effective organizational leadership and change skills for managing projects and project teams. CO 12 Recognize the need for life-long learning by undergoing the project work.
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